

F-distribution:

If X & Y are two chi square variate with df n_1 & n_2 then

$F = \frac{X/n_1}{Y/n_2}$ is called a

F-statistics.

and it follows a F distribution -

with pdf

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{n_1/2} F^{\frac{n_1}{2}-1}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right) \left[1 + \frac{n_1}{n_2} F\right]^{(n_1+n_2)/2}}$$

$$0 \leq F < \infty$$

where n_1 & n_2 are df

denoted as $F \sim F(n_1, n_2)$.

$$\mu'_r = E(F^r) = \int_0^\infty F^r f(F) dF$$

$$= \left(\frac{n_2}{n_1}\right)^r \frac{1}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} B\left(r + \frac{n_1}{2}, \frac{n_2}{2} - r\right)$$

, $n_2 > 2r$.

Relation between t & F

If a statistics t follows t -distribution with df n then t^2 follows F distribution.

with $(1, n)$ df , i.e. $t^2 \sim F(1, n)$.

Relation between F & χ^2

If $F(n_1, n_2)$ distribution of $n_2 \rightarrow \infty$, then $\chi^2 = n_1 F$ will follow χ^2 -distribution with degrees of freedom n_1 .

mgf of F distribution does not exist.